

HISABDO AI

DAY 13 — OME SHA / AI-ML FORECASTING MODULE

Complete MVP Methodology, Prediction Logic, Validation, .NET Code & Setup Guide

1. Module Objective

This module defines the Day 13 AI/ML forecasting work for HisabDo. The goal is to use 3–6 months of historical income and expense data to produce an explainable next-month forecast. The forecasting engine produces the numeric result; the LLM, if used later, only explains the verified result.

MVP flow:

Previous 3–6 Months Data → Trend Analysis → Income Forecast + Expense Forecast → Expected Saving → Expected Closing Balance

2. MVP Forecasting Methodology

For the first MVP, use a simple historical-baseline approach. It is easy to understand, deterministic, easy to validate, and suitable before introducing a complex ML model.

Component	MVP Method
History	Use 3–6 complete historical months where available
Income forecast	Average of usable historical monthly income
Expense forecast	Average of historical monthly expenses
Saving forecast	Expected Income – Expected Expense
Closing balance	Opening Balance + Expected Saving, only when trusted opening balance exists
Trend analysis	Compare recent months with earlier months
Confidence	Data-quality rules based on usable history/completeness
Advanced ML	Not required for first MVP; evaluate after baseline validation

3. Historical Income & Expense Analysis

1. Collect authenticated user's historical transactions.
2. Separate Income and Expense transactions.
3. Group transactions by calendar month.
4. Calculate monthly income.
5. Calculate monthly expense.
6. Calculate monthly saving = income – expense.
7. Identify missing/empty months.
8. Compare recent monthly values with earlier months.
9. Record unusual one-time values as limitations rather than silently deleting them.
10. Prepare a clean time-series dataset for forecasting.

4. Prediction Logic

Baseline formulas:

Metric	Formula
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Monthly Income	Sum of Income transactions in month
Monthly Expense	Sum of Expense transactions in month
Monthly Saving	Monthly Income – Monthly Expense
Expected Income	Average of usable historical monthly income
Expected Expense	Average of historical monthly expense
Expected Saving	Expected Income – Expected Expense
Expected Closing Balance	Opening Balance + Expected Saving

Example: if the usable historical monthly income values are 120,000, 130,000 and 125,000, the baseline expected income is their average. The same process is applied to expense. The resulting saving is calculated by the backend.

5. Why This MVP Approach?

- Explainable: the team can show exactly how a prediction was produced.
- Reproducible: the same input produces the same baseline result.
- Easy to validate against known historical months.
- Works without a large ML training dataset.
- Lower implementation risk for the first MVP.
- Creates a baseline for later comparison with regression/time-series models.

6. Income Forecast

Use the average of usable historical monthly income values. Zero-income months should not automatically be treated as normal income; they should be reviewed as possible missing/irregular data. The implementation therefore uses positive income observations for the basic income baseline.

- If no usable income exists, expected income is 0 and the result must be marked as limited/unavailable for meaningful forecasting.
- Highly irregular income must be flagged as a limitation.
- Do not ask the LLM to predict income.

7. Expense Forecast

Use the average monthly expense across the selected historical months. Zero-expense months remain part of the time series because they represent a real absence of recorded expenses, but the data-quality result should identify missing/empty history.

- Large one-time expenses may distort the average.
- Recurring expense signals can be used in a later refinement.
- Do not silently remove valid transactions.

8. Expected Savings

Expected Saving = Expected Income – Expected Expense.

- Positive value means the forecasted baseline has more income than expense.
- Negative value means forecasted expense exceeds forecasted income.
- This is a forecasted value, not a guarantee.

9. Expected Closing Balance

Expected Closing Balance = Trusted Opening Balance + Expected Income – Expected Expense.

- Only calculate this when a trusted opening/current balance is supplied by the backend.
- If opening balance is unavailable, return null rather than guessing.
- AI must never create an opening or closing balance.

10. Trend Analysis

Signal	Interpretation
Recent income rising	Income has increased in recent history; baseline may understate future income
Recent income falling	Income has decreased in recent history; baseline may overstate future income
Recent expenses rising	Recent spending is increasing; forecast risk is higher
Recent expenses falling	Recent spending is decreasing
Stable history	Baseline average is more representative
Missing months	Forecast reliability is reduced
One-time spike	May distort simple average; report as limitation

Trend analysis is initially diagnostic. Do not automatically apply an aggressive trend multiplier until validation shows it improves accuracy.

11. Evaluation & Accuracy Metrics

Use historical holdout validation. Pretend a known historical month is the future: use the months before it to generate a forecast, then compare forecast with the actual month.

Metric	Formula / Use
MAE	Average absolute difference between predicted and actual values
MAPE	Average absolute percentage error; avoid/handle cases where actual = 0
Bias / Mean Error	Average(predicted – actual), to detect systematic over/under prediction
Directional Accuracy	Whether forecast correctly identifies increase/decrease direction when applicable

Evaluate income and expense separately. Saving can then be evaluated from the resulting forecast values. Closing balance should only be evaluated when a reliable opening balance exists.

12. Suggested MVP Acceptance Targets

Do not claim a universal numeric accuracy target before testing real HisabDo data. First establish a baseline using sample/holdout datasets, record MAE/MAPE, and use the measured result as the MVP benchmark.

- Every forecast must be reproducible.
- Every forecast must identify the history used.
- Insufficient data must not be presented as high confidence.
- Actual-vs-predicted validation must be documented.
- Large errors must be investigated rather than hidden.

13. Insufficient Data Handling

Scenario	Expected Behavior
6 usable months	Normal baseline; potentially High quality if complete
3–5 usable months	Forecast allowed with Medium quality
<3 usable months	Return Low/Limited quality and limitations
No usable transactions	Return Unavailable forecast

No income observations	Expected income cannot be meaningfully forecast
No trusted opening balance	Closing balance = null
Missing month	Keep month in series and flag data-quality limitation
Irregular history	Return forecast with limitation; do not claim certainty

14. Assumptions

- Transaction amount is stored as a valid non-negative value.
- Transaction type reliably identifies Income vs Expense.
- Transaction date is valid and can be grouped by month.
- User authorization is enforced before historical data is retrieved.
- Historical months are comparable.
- Forecast horizon is one month for MVP.
- Historical average is a baseline, not a guaranteed prediction.
- Recurring expense information is supplementary and must be treated as evidence, not certainty.

15. Forecasting Limitations

- Average-based forecasting can be affected by one-time large transactions.
- Income may be irregular or seasonal.
- Missing historical transactions reduce reliability.
- Future unexpected expenses are not predictable from history alone.
- The baseline does not automatically understand external economic events.
- Forecast confidence is a data-quality indicator, not a probability guarantee.

16. Complete .NET Setup

11. Create a Day 13 folder/project in the existing HisabDo .NET solution; do not create a second financial database.
12. Add the CashFlowTransaction DTO or map the code to the existing transaction entity.
13. Implement ICashFlowTransactionRepository using the real HisabDo DbContext/repository.
14. Implement IOpeningBalanceProvider only if the real backend has a trusted balance source.
15. Register the service with dependency injection.
16. Create a controller endpoint: GET /api/ai/cash-flow-forecast/{userId}?months=6&horizonMonths=1.
17. Protect the endpoint with the existing authentication/authorization mechanism.
18. Never trust a caller-supplied userId without verifying that the authenticated user is permitted to access it.
19. Return the verified forecast DTO.
20. Pass only the verified DTO to Aarti's AI explanation layer.
21. Add unit tests for formulas, missing data, authorization and edge cases.
22. Run holdout validation before calling the MVP forecast accurate.

17. C# Service — Implementation Notes

The companion .cs file supplied with this document contains the reusable CashFlowForecastService, repository contract, opening-balance provider contract, monthly aggregation, baseline prediction, confidence logic and safe missing-data handling.

- Namespace: HisabDo.AI.CashFlowForecasting
- Service: CashFlowForecastService
- Main method: ForecastNextMonthAsync(...)
- Default history: 6 months

- Minimum requested history: 3 months
- Method label: HistoricalBaseline
- LLM dependency: none

18. API Example

GET /api/ai/cash-flow-forecast/{userId}?months=6&horizonMonths=1

```
{
  "userId": "USER_ID",
  "historyMonthsUsed": 6,
  "forecastHorizonMonths": 1,
  "expectedIncome": 150000.00,
  "expectedExpense": 108500.00,
  "expectedSaving": 41500.00,
  "openingBalance": 200000.00,
  "expectedClosingBalance": 241500.00,
  "confidence": "Medium",
  "method": "HistoricalBaseline",
  "limitations": [
    "Forecast is based on historical transaction behavior.",
    "Actual future income and expenses may differ from the forecast."
  ]
}
```

19. Sample Dataset for Validation

Month	Income	Expense	Saving
Jan	120000	80000	40000
Feb	130000	90000	40000
Mar	125000	85000	40000
Apr	135000	95000	40000
May	128000	92000	36000
Jun	132000	88000	44000

Using these six months, the baseline income and expense forecast can be calculated and then tested against a known historical holdout month. For production validation, use a larger set of representative datasets.

20. Validation Job — Step by Step

23. Prepare 3–6 months of historical sample data.
24. Select one known month as the validation/holdout month.
25. Use only earlier months as model input.
26. Run the forecasting service.
27. Record expected income and expense.
28. Compare them with the actual holdout month's income and expense.
29. Calculate MAE, MAPE where valid, and mean error.
30. Repeat across multiple holdout months/datasets.
31. Calculate average performance.
32. Document failure cases.
33. Only then decide whether the baseline is sufficient or an advanced ML model is justified.

21. Advanced ML — Future Phase

After the MVP baseline is validated, Omesha can compare more advanced approaches such as linear regression, time-series models or other suitable models. The advanced model should only replace the baseline if validation demonstrates a meaningful improvement and the additional complexity is justified.

The model still outputs verified forecast values to the backend. The LLM remains an explanation layer.

22. Team Handoff

Person	Deliverable
Taha	Approve methodology, assumptions, validation plan and acceptance
Jaffer	Connect service to real HisabDo data and API
Omesha	Validate baseline, metrics, datasets and model decision
Aarti	Explain verified forecast result; no numeric invention
Laiba	Display forecast, confidence, limitations and AI insight

23. Final Acceptance Checklist

- ☐ MVP methodology approved.
- ☐ 3–6 month history requirement implemented.
- ☐ Monthly income/expense aggregation validated.
- ☐ Expected income implemented.
- ☐ Expected expense implemented.
- ☐ Expected saving implemented.
- ☐ Closing balance handled only with trusted opening balance.
- ☐ Insufficient data handled.
- ☐ Confidence/data-quality logic documented.
- ☐ Holdout validation completed.
- ☐ MAE/MAPE/mean-error results recorded.
- ☐ Sample datasets tested.
- ☐ Assumptions documented.
- ☐ Limitations documented.
- ☐ User isolation verified.
- ☐ AI receives verified values only.
- ☐ Advanced ML decision deferred until baseline evidence exists.

24. Final Definition

Day 13 MVP forecasting uses a simple, explainable historical baseline to estimate next-month income and expenses from available 3–6 month history. Expected saving is derived from those verified forecast values, and expected closing balance is calculated only when a trusted opening balance is available. The approach is validated through historical holdout testing and objective error metrics. AI may explain the verified forecast but cannot invent, change or recalculate financial numbers.